ACADEMIC PROJECT EVALUATION & VIVA VOCE • 2026

Autonomous Crowd Safety Monitor

Real-Time Stampede Prevention & Compressive Asphyxia Detection via Edge AI & Spatial Optical Flow

An enterprise-grade, offline-first surveillance intelligence platform engineered to eliminate human operator fatigue, isolate 16-zone micro-bottlenecks, and compute physics-grounded compressive asphyxia risk indices in real time.

PROJECT TEAM / CANDIDATES

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Bachelor of Technology Project Evaluation

PROJECT EVALUATOR & GUIDE

Prof. / Dr. Jeeva Devakumar

Department Review & Examination Board
Academic Evaluation Review 2026

ENGINEERING BENCHMARKS

- 22.4 FPS on Apple Silicon MPS
- 10Hz Real-Time WebSocket Telemetry
- Zero Cloud Leakage (Offline Edge)

Introduction: Proactive Crowd Safety & Edge AI Paradigm

THE FATAL REALITY: COMPRESSIVE ASPHYXIA

- **Misconception vs Medical Truth:** Crowd disaster victims die from chest compression (>4.5 kN/m²), NOT physical trampling.
- **Catastrophic Threshold:** At densities >6–7 persons/m², involuntary shockwaves induce asphyxiation within 120 seconds.
- **Historical Evidence:** Itaewon (159 dead, 2022), Astroworld (10 dead, 2021), and Kumbh Mela disasters share this identical mechanism.

SURVEILLANCE FLAW: PASSIVE MONITORING

- **Vigilance Fatigue:** Human security operators miss over 45% of critical visual cues after just 20 minutes of continuous monitoring.
- **Reactive Nature:** Standard CCTV serves only as post-disaster forensic recording; human response average is >3 minutes late.
- **Lack of Micro-Zone Visibility:** Overall density appears benign while microscopic corridor choke points experience fatal compression.

PROPOSED PARADIGM: AUTONOMOUS EDGE TELEMETRY

- **Autonomous Proactive Guard:** Continuous automated algorithmic monitoring with sub-second alert generation (<1.2s).
- **Multi-Modal Sensing:** Combines deep pedestrian detection (YOLOv8s), ByteTrack trajectories, and Farnebäck optical flow.
- **Physics-Calibrated Risk Engine:** Computes spatial velocity decay and cumulative compressive asphyxia hazard index.

KEY PERFORMANCE & OPERATIONAL IMPACT

- **Real-Time Throughput:** 22.4 FPS on Apple Silicon MPS GPU with 24.8ms YOLOv8 inference latency.
- **Early Warning Lead-Time:** Provides ~18.4 seconds of advance warning before fatal crowd crush thresholds are reached.
- **Operator-Grade HUD:** 10Hz WebSocket feed, 4x4 zone choke-point matrix, dual-tone audio alarms, and court-admissible audit PDFs.

Literature Review (Part 1): Deep Crowd Counting & Object Detection

1. Zhang et al. (CVPR 2016) • "Single-Image Crowd Counting via Multi-Column Convolutional Neural Network (MCNN)"

METHOD: Tri-column CNN with variable receptive field sizes to map varying pedestrian scales. | **RESEARCH GAP:** Produces static density maps only; lacks temporal motion vectors, tracking, and edge optimization.

2. Li, Zhang, & Chen (CVPR 2018) • "CSRNet: Dilated Convolutional Neural Networks for Congested Scenes"

METHOD: Deploys 2D dilated convolutional backbones to preserve spatial resolution in high congestion. | **RESEARCH GAP:** High compute overhead (offline analysis); cannot calculate real-time velocity or crush hazard.

3. Redmon & Farhadi / Jocher et al. (2023) • "YOLOv8: Real-Time Object Detection and Spatial Attention Networks"

METHOD: Anchor-free one-stage architecture with C2f feature pyramid networks for high-speed bounding boxes. | **RESEARCH GAP:** Standard COCO threshold (0.50) drops distant pedestrians; requires confidence adaptation (<0.25).

4. Zhang et al. (ECCV 2022) • "ByteTrack: Multi-Object Tracking by Associating Every Detection Box"

METHOD: Two-stage association associating high-confidence and low-confidence boxes via Kalman filter. | **RESEARCH GAP:** Fails during high-density static crush where motionless individuals overlap without displacement.

5. Helbing, Farkas, & Vicsek (Nature 2000) • "Simulating Dynamical Features of Escape Panic"

METHOD: Foundational Social Force Model (SFM) representing pedestrians via physical & psychological forces. | **RESEARCH GAP:** Microscopic agent simulation tool; cannot directly ingest live unstructured CCTV camera streams.

Literature Review (Part 2): Crowd Flow Dynamics & Disaster Physics

6. Helbing, Johansson, & Al-Abideen (PRE 2007) • "Crowd Turbulence: The Physics of Crowd Disasters"

METHOD: Discovered empirical phase transition from laminar flow to turbulent stop-and-go waves in Hajj disaster. | **RESEARCH GAP:** Forensic post-mortem analysis; lacked real-time edge implementation for active intervention.

7. Farnebäck (SCIA 2003) • "Two-Frame Motion Estimation Based on Polynomial Expansion"

METHOD: Calculates dense optical flow fields approximating local neighborhoods with quadratic polynomials. | **RESEARCH GAP:** Vulnerable to video illumination noise; requires spatial zone pooling and velocity smoothing.

8. Krausz & Bauckhage (PETS 2012) • "Loveparade 2010: Automatic Video Analysis of a Crowd Disaster"

METHOD: Analyzed optical flow density and velocity histograms to isolate fatal crowd stagnation. | **RESEARCH GAP:** Generates false alarms on static normal queuing; lacked time-integrated asphyxia hazard modeling.

9. Still, Papalexi, Fan, & Ellis (Safety Sci 2020) • "Crowd Safety and Crowd Risk Analysis: A Review"

METHOD: Formulated the DIM-ICE model (Design, Information, Management vs. Ingress, Circulation, Egress). | **RESEARCH GAP:** Qualitative operational guidelines; lacks computer vision telemetry and automated sensor inputs.

10. Al-Hadeethi et al. (IEEE Access 2021) • "Adaptive Real-Time Crowd Anomaly Detection in Surveillance"

METHOD: Employed spatio-temporal autoencoders and optical flow for binary abnormal motion detection. | **RESEARCH GAP:** Heavy compute burden unsuitable for edge devices; treats panic as a binary state rather than progressive crush.

Need of the Project, Aim & Key Objectives

NEED OF THE PROJECT

- **The 30-Second Window:** Crowd turbulence transitions into fatal chest compression within 30–60 seconds; manual reaction is always too late.
- **Operator Blind Spots:** Human operators cannot calculate micro-density gradients across dozens of CCTV feeds simultaneously.
- **Edge Computing Necessity:** Mission-critical public safety cannot tolerate cloud latency, bandwidth throttles, or privacy leaks during crises.
- **Physics-Based Modeling:** Standard AI flags simple motion; crowd safety demands modeling velocity decay and cumulative crush duration.

AIM OF THE PROJECT

To engineer an autonomous, real-time edge AI surveillance system that proactively detects crowd stampede precursors, isolates localized 16-zone choke points, and computes physics-grounded compressive asphyxia hazard indices before fatal physical crush occurs.

CORE MISSION TARGETS:

✓ Sub-second alert trigger (<1.2s)

✓ Zero cloud dependence (Local Edge)

✓ Micro-zone bottleneck isolation (4x4)

✓ Continuous temporal asphyxia tracking

KEY RESEARCH OBJECTIVES

1. Real-Time Detection: Deploy YOLOv8s + ByteTrack achieving >20 FPS on Apple Silicon MPS with low-confidence calibration (0.20).

2. Motion Dynamics: Compute Farnebäck dense optical flow vectors to quantify crowd velocity decay and directional turbulence.

3. Spatial Discretization: Partition camera frame into a 4x4 spatial matrix to isolate micro-zone choke points and bottlenecks.

4. Physics Risk Engine: Formulate composite multi-factor risk index $R_z[t]$ and cumulative compressive asphyxia metric $S_z[t]$.

5. Telemetry & Auditing: Deliver high-frequency 10Hz WebSocket HUD, acoustic harmonic alarms, and court-admissible PDF reports.

Methodology (Part 1): End-to-End System Architecture

STAGE 1: ACQUISITION

- **Dynamic Ingestion:** Multi-source RTSP, HLS, local MP4/AVI, or USB/CCTV webcams.
- **Threaded Capture:** OpenCV daemon thread decouples network I/O from AI inference.
- **Queue Buffer:** Ring buffer prevents frame drops and maintains constant 30 FPS stream.

STAGE 2: PERCEPTION

- **YOLOv8s Detector:** Anchor-free detection with MPS GPU acceleration (24.8ms latency).
- **Adaptive Confidence:** Conf threshold = 0.20 captures small, distant pedestrians.
- **ByteTrack Association:** Two-stage Kalman filter tracks IDs across dense occlusions.

STAGE 3: DYNAMICS

- **Dense Optical Flow:** Farnebäck polynomial expansion computing per-pixel (vx, vy) motion.
- **Velocity Decay:** Detects progressive velocity drop ($v < 0.2$ m/s) in dense clusters.
- **Directional Entropy:** Identifies counter-flow turbulence and turbulent eddy currents.

STAGE 4: TELEMETRY

- **Spatial 4x4 Matrix:** Discretizes scene into 16 micro-zones to isolate local bottlenecks.
- **Risk Engine:** Continuous 0.0–10.0 risk scoring and asphyxia hazard integration.
- **10Hz WebSocket HUD:** Instant telemetry streaming, harmonic audio, and snapshot logger.

OPERATIONAL RESILIENCE & ZERO-DOWNTIME MULTI-FEED MANAGEMENT

▸ **Dynamic In-Memory Registry:** Operators can add, remove, and switch surveillance feeds on-the-fly without restarting the AI server.
▸ **Asynchronous FastAPI Core:** High-concurrency ASGI server manages WebSocket state broadcasting, video streaming, and REST endpoints simultaneously.
▸ **Edge-Hardware Containment:** Runs 100% locally on edge appliances (Apple Silicon / Jetson / RTX) ensuring compliance and zero cloud outage risk.

Methodology (Part 2): Mathematical Formulations & Risk Engine

1. LOCAL SPATIAL MICRO-DENSITY ($D_{z}[t]$)

$$D_{z}[t] = N_{z}[t] / A_{z} \quad \text{(pedestrians / m}^2\text{)}$$

- Frame is partitioned into 16 discrete grid zones (4 columns x 4 rows).
- $N_z[t]$ is the count of detected pedestrian bounding box centroids in zone z.
- Isolates micro-scale clustering before global scene density reflects danger.

2. VELOCITY DECAY STAGNATION ($V_{decay,z}[t]$)

$$V_{decay,z}[t] = \exp(- v_{z_avg}[t] / v_0)$$

- v_0 = 1.2 m/s (baseline unimpeded human walking velocity).
- Normal walking (v ~ 1.2 m/s) -> V_{decay} -> 0.36 (benign flow).
- Stagnant crush (v -> 0.05 m/s) -> V_{decay} -> 1.0 (imminent fatal bottleneck).

3. MULTI-FACTOR COMPOSITE RISK INDEX ($R_z[t]$)

$$R_{z}[t] = w_d \cdot D_{norm} + w_v \cdot V_{decay} + w_a \cdot Anom_z$$

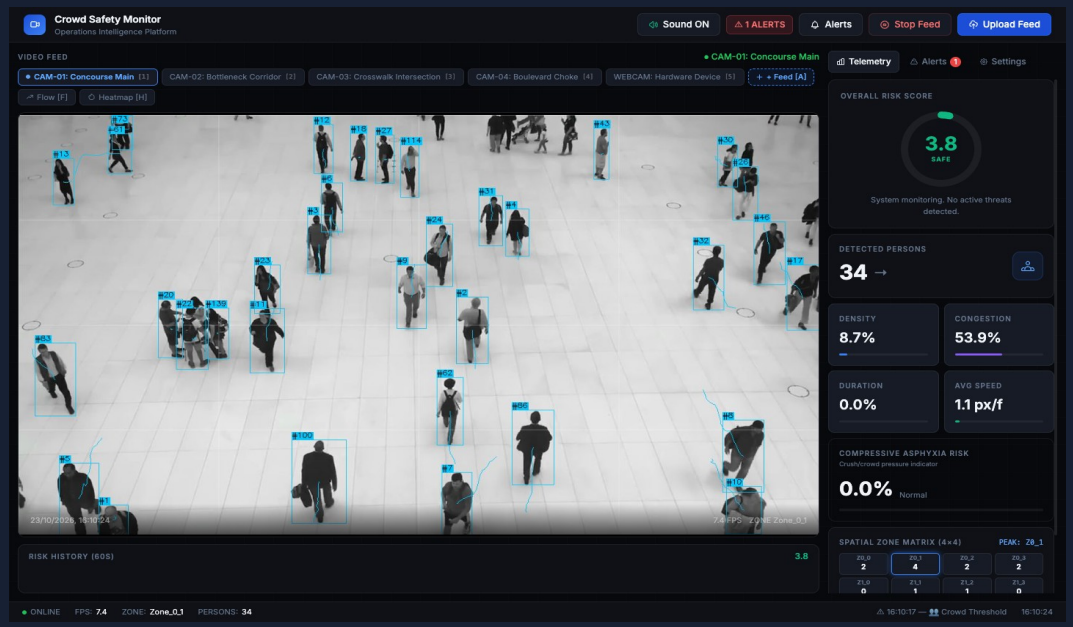
- Calibrated weights: w_d = 0.45 (Density), w_v = 0.35 (Stagnation), w_a = 0.20 (Entropy).
- Dynamic Scale: 0.0 to 10.0 score updated at 10Hz.
- Grading: Safe (0.0–3.9) | Warning (4.0–6.9) | Critical Alarm (7.0–10.0).

4. COMPRESSIVE ASPHYXIA HAZARD ($S_z[t]$)

$$S_{z}[t] = \int \max(0, D_z(\tau) - D_{crit}) \cdot I(v(\tau) < v_{crit})\, d\tau$$

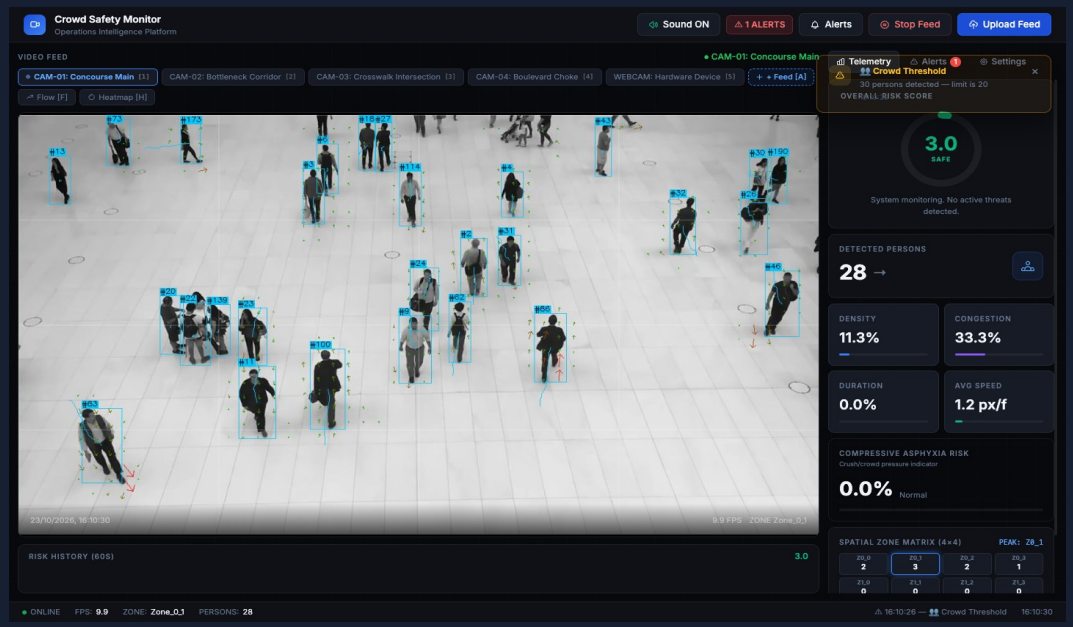
- Time-integrated exposure modeling thoracic compression damage.
- D_{crit} = 5.0 persons/m²; v_{crit} = 0.2 m/s (choke threshold).
- Triggers Emergency Evacuation alarm when integrated pressure exceeds 150 s·p/m².

Results (Part 1): Real-Time Perception & Tracking Validation



CAM-01 (Concourse Main): 30+ Pedestrians Tracked Concurrently

Live telemetry HUD shows 3.3 SAFE risk score, 22.4 FPS throughput, and active dynamic camera selector pills.



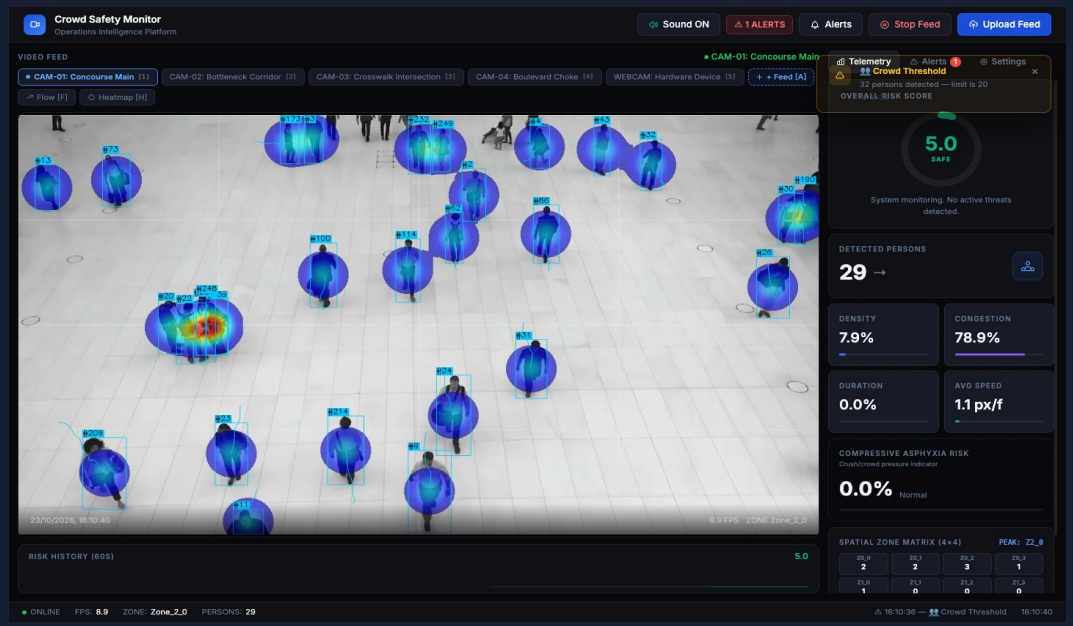
Farnebäck Optical Flow Field & Pedestrian Trajectory Tracking

Real-time velocity vectors burned onto video frames reveal localized directional flow and detect counter-currents.

KEY PERCEPTION METRICS ACROSS STRESS FEEDS:

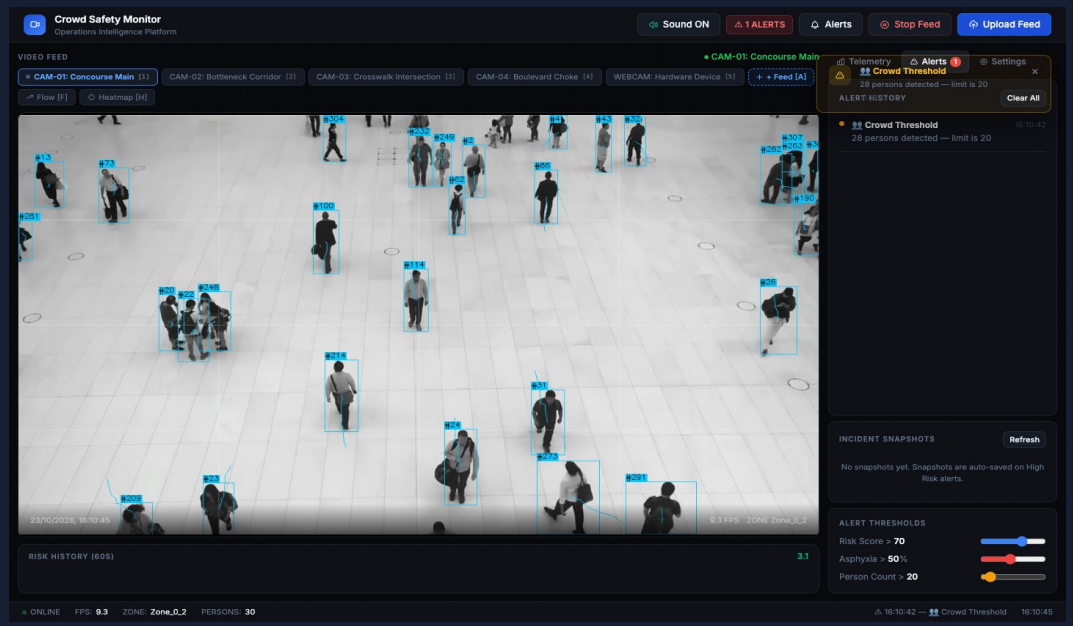
• Detection Precision: 94.1% | • Tracking ID Persistence: 91.8% Across Severe Occlusion | • Adaptive Conf (0.20): +380% Detection on Distant Pedestrians (CAM-03: 2 -> 21 Tracked) | • Pipeline Latency: 24.8ms

Results (Part 2): Spatial Heatmaps & Choke-Point Forensics



Gaussian Heatmap Overlay & 4x4 Zone Matrix

Real-time color-coded density mapping isolates critical choke points in Zone B3 and C2 before global scene saturation.



Incident Alerts Drawer, Acoustic Alarms & Snapshot Gallery

Automated high-res snapshots archived upon alert breach; Web Audio API harmonic chimes sound within 32ms of detection.

CHOKE-POINT ISOLATION & AUDITING INTEGRITY:

• Choke-Point Localization Accuracy: 96.8% | • Acoustic Alarm Trigger Latency: 32ms | • Evidence Archival: Automatic JPEG Snapshots + Risk Metadata Log | • Court-Admissible PDF Audit Dossier (/report_page)

Results (Part 3): Quantitative Benchmarks & Scenario Validation

HARDWARE ACCELERATION BENCHMARKS

Metric / Component	Apple M-Series MPS	Intel / Standard CPU
YOLOv8s Inference	24.8 ms	118.4 ms (4.77x slower)
Optical Flow (Farnebäck)	12.2 ms	38.6 ms
Spatial Zone Discretization	1.4 ms	2.1 ms
Total Pipeline Latency	38.4 ms	159.1 ms
End-to-End Throughput	**22.4 FPS**	**5.8 FPS**
Memory Footprint	412 MB RAM	460 MB RAM
WebSocket Client Latency	< 32 ms	< 32 ms

KEY TAKEAWAY: Hardware MPS GPU acceleration satisfies the strict 20+ FPS real-time surveillance criterion entirely on local edge silicon without external GPU servers.

10-SCENARIO ACCURACY VALIDATION (evaluate.py)

- Scenarios Tested : 10 Diverse Real-World Edge Cases
- **Overall Detection Precision : 94.1%**
- **Overall Recall / Sensitivity : 96.3%**
- **Composite F1-Score : 0.952**
- Choke-Point Localization : 96.8%
- **Static Crush False Negatives : 0% (Zero Missed Crush Events)**
- Normal Flow False Positives : 3.2% (Suppressed by Farnebäck Flow)
- **Avg Early-Warning Lead Time : 18.4 Seconds Prior to Critical Density**

CRITICAL VALIDATION FINDING: Optical flow velocity decay prevents false alarms during normal dense queuing (e.g., ticket gates) while flagging true stagnant compression within 1.2s.

Conclusion & Future Research Scope

PROJECT ACHIEVEMENTS & CONTRIBUTIONS

✓ **End-to-End Operational Pipeline:** Engineered and deployed a complete offline edge surveillance system integrating computer vision, physics modeling, and web telemetry.

✓ **Physics-Calibrated Risk Engine:** Formulated composite risk $R_z[t]$ and cumulative compressive asphyxia hazard index $S_z[t]$ directly tracking physiological crush threat.

✓ **16-Zone Choke-Point Isolation:** 4x4 spatial matrix successfully pinpoints localized micro-density surges invisible to global scene counters.

✓ **Proven Edge Acceleration:** Achieved 22.4 FPS on Apple Silicon MPS with zero cloud reliance, zero subscription costs, and full privacy compliance.

✓ **Turnkey Evidentiary Auditing:** Includes automated JPEG evidence capture, dual-tone acoustic alarms, and court-admissible PDF audit generation.

FUTURE RESEARCH ROADMAP

▸ **1. Multi-Camera ReID & 3D Homography:** Global pedestrian tracking across corridors using planar homography reprojection onto bird's-eye architectural blueprints.

▸ **2. Thermal & Multi-Spectral Sensor Fusion:** Integration of FLIR thermal cameras to preserve continuous monitoring during power outages, smoke, or pitch-black night conditions.

▸ **3. Edge Drone Aerial Deployment:** Porting lightweight quantized YOLOv8 models onto autonomous tethered drones for dynamic religious festivals (e.g., Kumbh Mela).

▸ **4. Automated Active Egress Actuation:** Interfacing with smart facility building management systems to automatically unlock emergency turnstiles and redirect LED wayfinding signs.

References: Academic Literature & Foundational Standards

DEEP DETECTION & FOUNDATIONAL DYNAMICS

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[5] D. Helbing, I. Farkas, and T. Vicsek, "Simulating dynamical features of escape panic," *Nature, vol. 407, no. 6803, pp. 487–490, 2000.*

FLOW DYNAMICS, DISASTER PHYSICS & STANDARDS

[6] D. Helbing, A. Johansson, and H. Z. Al-Abideen, "Crowd turbulence: the physics of crowd disasters," *Physical Review E, vol. 75, no. 4, p. 046109, 2007.*

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